Week 3 Lab: VAE, Latent Diffusion and Flow Matching

Exercise 1

The Inception Score (IS) feeds each generated image y^n into an off-the-shelf classifier to get $P(c|y^n)$, and averages these to get $P(c) = \frac{1}{N}\sum_n P(c|y^n)$. Formally,

$$\text{IS} = \exp\Big(\frac{1}{N}\sum_n \text{KL}(P(c|y^n) \,\|\, P(c))\Big).$$

(a) According to the slides, what should $P(c|y)$ look like for a high-*quality* image, and what should $P(c)$ look like for a *diverse* set?
(b) Compute IS for three images with classifier outputs (0.9, 0.05, 0.05), (0.05, 0.9, 0.05) and (0.05, 0.05, 0.9). Then recompute when the second image is replaced by a copy of the first. Interpret.
(c) Name one failure mode that IS cannot detect.

Solution

(a) Quality: $P(c|y)$ concentrated on one class (the classifier is confident). Diversity: $P(c)$ close to uniform (all classes appear). High IS requires both.
(b) Set 1: $P(c) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; each KL $= 0.9 \ln 2.7 + 2 \cdot 0.05 \ln 0.15 \approx 0.704$, so IS $= e^{0.704} \approx 2.02$ (the maximum possible here is 3, the number of classes). Set 2: $P(c) = (0.617, 0.05, 0.333)$, mean KL ≈ 0.420, IS ≈ 1.52. Fewer distinct classes means lower diversity and a lower score (a mild form of mode collapse/dropping, pp. 5–6).
(c) IS never looks at real data. It cannot detect memorisation (copying or flipping training images scores perfectly, p. 10), diversity *within* a class (it only sees class labels), or images that are unrealistic but classified confidently.

Exercise 2

FID embeds real and generated images with the Inception network, fits a Gaussian to each set of embeddings and reports the Fréchet distance between the two Gaussians. In 1-D this distance is $(\mu_r - \mu_g)^2 + (\sigma_r - \sigma_g)^2$.

(a) Real features are $\mathcal{N}(0, 1)$. Compute the 1-D FID for generated features $\mathcal{N}(0.5, 1)$, $\mathcal{N}(0, 0.25)$ (std 0.5), and $\mathcal{N}(0, 1)$.
(b) Which of these corresponds to mode collapse? Is a smaller or a larger FID better?
(c) Why does FID need "a lot of samples", and why is it still not enough to detect a model that simply copies training images?

Solution

(a) $\mathcal{N}(0.5, 1)$: $0.25 + 0 = 0.25$. $\mathcal{N}(0, 0.25)$: $0 + (1 - 0.5)^2 = 0.25$. $\mathcal{N}(0, 1)$: 0.
(b) The narrow distribution (std 0.5) is mode collapse: samples are less varied than real data. Smaller FID is better. Note that a shift in quality/content and a loss of diversity can give the same FID; the single number mixes both.
(c) Means and (in high dimension, e.g. 2048-D) covariance matrices must be estimated from samples. With few samples the estimates are noisy and FID is biased upward, so thousands of images (commonly 10k–50k) are used. A model that outputs copies of the training set has the same feature statistics as the real data, so FID is near-perfect even though nothing new is generated (p. 10).

Exercise 3

For text-to-image models we also want to know whether the image matches the prompt ("A running dog").

(a) How is CLIP trained on its 400 million image–text pairs, and how is it used to score a generated image?
(b) Give one situation in which a high CLIP score does not mean a good image.

Solution

(a) An image encoder and a text encoder are trained *contrastively*. In each batch, matching image–caption pairs should have high embedding similarity and all mismatched pairs low similarity. To evaluate, embed the prompt and the generated image and use their cosine similarity (CLIP score); higher means better agreement.
(b) CLIP judges semantic agreement, not visual quality. An image with the right content but artefacts can score well. CLIP is also weak at counting, spatial relations and attribute binding ("red cube on a blue sphere"), and it can be fooled by writing the prompt text into the image. Optimising a generator directly against CLIP can exploit these weaknesses.

Exercise 4

In the lecture's VAE the encoder outputs m_i and σ_i, samples $e_i \sim \mathcal{N}(0,1)$, and forms the code $c_i = \exp(\sigma_i)\, e_i + m_i$. Besides the reconstruction error it minimises $\sum_i \left(\exp(\sigma_i) - (1+\sigma_i) + m_i^2\right)$.

(a) Why does decoding a *randomly generated* code usually fail for a plain auto-encoder?
(b) What happens if the VAE only minimises the reconstruction error? (Consider what the encoder would do with σ_i.)
(c) Show that the extra term is minimised at $\sigma_i = 0$, $m_i = 0$. What does this mean for the distribution of codes? Which part acts like L2 regularisation?

Solution

(a) The AE is only trained on codes produced by its encoder, which may occupy a small, irregular region of latent space. A random code usually lands where the decoder was never trained, so its output is meaningless (Lab 2A: the AE grid decodes well only near encoded points).
(b) The noise makes reconstruction harder, so the encoder would drive $\sigma_i \to -\infty$ (noise std $\exp(\sigma_i) \to 0$). The VAE then degenerates into a plain AE.
(c) Derivative w.r.t. σ_i: $\exp(\sigma_i) - 1 = 0 \Rightarrow \sigma_i = 0$ (second derivative $e^{\sigma_i} > 0$, so a minimum); m_i^2 is minimised at $m_i = 0$. The minimum value is $1 - 1 + 0 = 0$. It pushes each code distribution toward $\mathcal{N}(0,1)$: noise std near 1 and means near 0, so the codes fill the region we will later sample from. The m_i^2 term is exactly an L2 penalty on the code means. (It is the per-dimension KL to $\mathcal{N}(0,1)$ up to a factor 2, with σ_i playing the role of a log-variance, as in the lab's $\frac{1}{2}\sum(\mu^2 + e^{\text{logvar}} - 1 - \text{logvar})$.)

Exercise 5

In the lab both models have a 2-D latent space; you plot the codes of held-out digits and decode a grid inside a dashed square (the region where most mass of $\mathcal{N}(0, I)$ lies).

(a) Where would you expect AE codes and VAE posterior means to lie relative to the square, and where does each decoded grid produce recognisable digits?
(b) Must the VAE's KL curve decrease every epoch? Explain the trade-off between the two terms.

Solution

(a) AE codes can sit anywhere, often stretched far outside the square or clumped in a few regions with empty gaps. Its grid gives digits only near encoded points and garbage elsewhere. VAE means are pulled toward the origin and spread more evenly inside the square, so most grid points decode to plausible digits with smooth transitions between classes. (VAE means need not themselves look exactly Gaussian; the KL acts on the full posteriors.)
(b) No. Early in training the KL is often small because the encoder carries little information about the input. As the model learns to encode more information for reconstruction, the KL can *rise*. The objective balances reconstruction (use the latent) against staying close to the prior, so only the total loss is expected to fall.

Exercise 6

Flow-matching training pairs a noise point $x_0 \sim \mathcal{N}(0, I)$ with a data point x_1, samples $t \sim U(0, 1)$, forms $x_t = (1 - t)x_0 + tx_1$, and regresses the network output $\hat{v} = v_\theta(x_t, t)$ onto $v = x_1 - x_0$ with MSE.

(a) For $x_0 = (1, -1)$, $x_1 = (3, 2)$ and $t = 0.25$, compute the network input x_t and the target v. Show that $v = \frac{d}{dt}x_t$.
(b) Two different pairs can pass through (almost) the same (x_t, t) with different targets v. What does the MSE-optimal network predict there? Are the learned sampling trajectories straight?
(c) Compare with the diffusion training loop (L3 p. 39): what is the network asked to predict in each case?

Solution

(a) $x_t = 0.75(1, -1)+0.25(3, 2) = (1.5, -0.25)$; $v = (3, 2)-(1, -1) = (2, 3)$. $\frac{d}{dt}[(1-t)x_0+tx_1] = x_1 - x_0$: the straight line between the pair is travelled at constant velocity.
(b) The MSE minimiser is the *conditional average* of the targets, $v^*(x, t) = \mathbb{E}[x_1 - x_0 \mid x_t = x]$: an average over all pairs whose paths cross there. Averages of crossing straight lines point in compromise directions, so the learned (marginal) trajectories are generally curved, even though each training path is straight. (Minibatch optimal-transport couplings, p. 41, reduce crossings and straighten the flow.)
(c) Diffusion: the noise ε that was added ($\|\varepsilon - \hat{\varepsilon}\|^2$). Flow matching: the velocity $x_1 - x_0$ that moves the sample from noise toward data. Both use an MSE regression on a synthetic noisy input; only the target and the way the prediction updates the sample differ.